

## Performance Testing

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 06 July 2026                                                |
| Team ID       | SWTID-2026-4324                                             |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                     |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field               | Details                                                      |
|---------------------|--------------------------------------------------------------|
| Testing Tool Used   | Apache JMeter                                                |
| Type of Testing     | Load Testing & Performance Testing                           |
| Target Module / API | Crop Recommendation, Weather Forecast, User Login, Dashboard |
| Test Environment    | Local System (Windows 11, XAMPP, MySQL, Python/Flask)        |
| Test Date           | 06 July 2026                                                 |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome                                    |
|------|-----------------------------|----------------------|----------------|-----------------------------------------------------|
| 1    | User Login Request          | 20                   | 60             | Login should complete successfully within 2 seconds |
| 2    | Crop Recommendation Request | 50                   | 120            | Accurate crop recommendation with quick response    |
| 3    | Weather Data Retrieval      | 40                   | 60             | Weather data displayed without errors               |

|   |                   |    |    |                                          |
|---|-------------------|----|----|------------------------------------------|
| 4 | Dashboard Loading | 30 | 90 | Dashboard loads smoothly with all charts |
|---|-------------------|----|----|------------------------------------------|



### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                 |
|------|----------------------|--------------|--------------|----------------------|-------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.3 sec      | Pass                 | Meets target            |
| 2    | Response Time (Max)  | < 5 seconds  | 3.6 sec      | Pass                 | Stable under load       |
| 3    | Throughput (Req/sec) | > 40 Req/sec | 52 Req/sec   | Pass                 | Good performance        |
| 4    | Error Rate           | < 1%         | 0.2%         | Pass                 | Very low errors         |
| 5    | CPU Utilization      | < 80%        | 64%          | Pass                 | Within acceptable limit |
| 6    | Memory Utilization   | < 80%        | 68%          | Pass                 | Stable memory usage     |

### Step 4: Observations & Analysis

#### Key Findings:

- The OptiCrop system handled multiple users efficiently.
- Response time remained below the target value during testing.
- Crop recommendation and weather modules performed reliably.
- No major performance degradation was observed.

#### Bottlenecks Identified:

- Slight increase in response time during simultaneous weather API requests.
- Minor delay while loading dashboard graphs under high load.

#### Optimization Steps Taken:

- Optimized database queries.
- Reduced API response time using caching.
- Improved backend processing logic.
- Optimized dashboard data loading.

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]